

Global Energy Decision Atlas Data and Methodology Note

Purpose

The atlas screens national markets before local energy diligence. It keeps electricity price, generation structure, primary energy scale, and energy balance as separate measures so a reviewer can see the tradeoffs behind a shortlist.

Two datasets

Dataset	Coverage and periods	Balance measure
Assignment 15	15 countries. December 2025 price, 2024 consumption and mix, 2023 trade.	Reported total energy net imports.
Expanded 50	50 countries. 2023-2026 average price and 2023 energy measures.	Primary energy consumption minus total energy production, used as a screening proxy.

Indicators and sources

Indicator	Definition and unit	Primary source
Electricity price	Residential average use or commercial 1,000,000 kWh per year, USD/kWh.	GlobalPetrolPrices public comparison.
Non-fossil share	Solar, wind, hydro, and other share of domestic generation, percent.	Our World in Data Energy CSV.
Energy scale	Total primary energy consumption, EJ.	Assignment sources or Our World in Data.
Trade or balance	Reported net imports for Assignment 15; derived balance gap for Expanded 50, EJ.	Assignment source or EIA International.

Coverage and aggregation

Expanded 50 covers household price in 47/50 markets and business price in 45/50. Consumption, electricity mix, and the balance gap each cover 47/50. A fallback may use a value up to two years earlier and remains flagged. Missing values stay N/A. Regional price uses the unweighted member median. Consumption and balance sum available member values. Regional electricity mix weights member shares by domestic generation. The interface reports overlapping region membership and calculates Pearson r from at least five complete pairs.

Analysis maturity

Level	Status	What this work supports	What it does not prove
Descriptive	performed	Coverage, extrema, rankings, concentration, complete-case correlations, and fixed comparisons.	Causation, future outcomes, or an optimal market.
Diagnostic	partial	Associations and external context can form hypotheses and local questions.	Why an observed price, mix, demand, reliability, or balance value occurred.
Predictive	not performed	The snapshot establishes a baseline for later forecast design.	Future electricity prices, demand, reliability, or emissions.
Prescriptive	not performed	A diligence shortlist can order follow-up work.	A best market, investment return, or optimal site.

Limits and decision use

The market scope is non-random. Missing values, mixed reference years, flagged fallbacks, national averages, and source-definition differences limit comparison. Correlations use complete cases and have no confidence intervals. The Expanded 50 balance gap is not an observed trade flow. Use the shortlist to plan diligence, not to approve an investment. A site decision needs current industrial tariffs, capital cost, contracts, connection capacity, reliability records, regulatory constraints, and an explicit objective and weighting.